A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Figure 2: Truth table of AND gate

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\0\end{bmatrix}+\begin{bmatrix}0\\1\end{bmatrix}\right) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Figure 3: Hadamard matrix operation on $|0\rangle$

Now let us turn our attention to quantum gates. Before proceeding further, it is important to note that quantum gates are fundamentally different from the classical logic gates used in present-day digital computers. In simple terms, classical gates are not reversible, whereas quantum gates are reversible. To understand this difference, let us consider a classical gate widely used in digital computers: the AND gate. Figure 2 shows the truth table of AND gate with two input bits.

From the figure it is easy to see that the operation of the AND gate cannot be reversed. For example, if the output of an AND gate is 0, there is no way to uniquely determine the exact values of the input bits A and B. The only conclusion we can draw is that the case A=1 and B=1 did not occur. A quantum gate, on the other hand, is reversible. If the same quantum gate is applied twice, we obtain the original state again. This is possible because a quantum gate performs a reversible transformation of the quantum state, typically represented mathematically by a unitary matrix acting on the qubit. At this point, I am sure that the previous sentence may have sounded a bit intimidating and might even have dampened your enthusiasm for learning quantum computing. But do not worry—we will gradually unpack ideas like these in the coming articles in this series. By the end of the series, statements like this will no longer seem mysterious or frightening.

Let us now consider one of the most important quantum gates called the Hadamard gate and study its effect on a qubit. The Hadamard gate is named after the famous French mathematician Jacques Hadamard, who studied the properties of Hadamard matrices. Let us see what happens when a Hadamard gate is applied to a qubit in the state $|0\rangle$. As discussed earlier, the state $|0\rangle$ can be represented by the vector $[1 \quad 0]^T$. Similarly, the state $|1\rangle$ can be represented by the vector $[0 \quad 1]^T$. A Hadamard gate acting on a single qubit is represented by a special matrix called the Hadamard matrix, denoted by H. The Hadamard matrix H is shown in Figure 3. In this article, we will focus only on the mathematical operation that occurs when a Hadamard gate is applied to a qubit in the state $|0\rangle$.

From Figure 3, we can see that after applying the Hadamard gate, the value of the qubit changes from $|0\rangle$ to $\sqrt{(1/2)}|0\rangle+\sqrt{(1/2)}|1\rangle$. Notice that the only mathematical operations involved in the calculations shown in Figure 3 are matrix multiplication and matrix addition. Recall that we previously established that a qubit in the superposition state $|\psi\rangle=\sqrt{(1/2)}(|0\rangle+|1\rangle)$ has an equal probability of being measured as $|0\rangle$ or $|1\rangle$. Thus, the Hadamard gate serves as one of the most important building blocks of quantum circuits, as it creates superposition.

Now let us examine the physical interpretation of applying a Hadamard gate using the Bloch sphere representation. From Figure 1, we know that a qubit in the state $|0\rangle$ is represented by an arrow pointing towards the north pole of the Bloch sphere. Let us again use the Bloch sphere simulator mentioned earlier to observe the effect of the Hadamard gate.

In the Bloch sphere simulator web page, there is a menu labelled 'Quantum gates', which includes an option to apply the Hadamard gate. When the Hadamard gate is applied, the Bloch sphere configuration shown in Figure 4 is obtained. From Figure 4, we can see that the state of the qubit now points toward the positive x-axis. It can also be observed that the qubit state is now equidistant from the north and south poles of the Bloch sphere. This observation confirms our earlier calculation: a qubit in the state $|\psi\rangle=\sqrt{(1/2)}(|0\rangle+|1\rangle)$ will produce the outcomes $|0\rangle$ or $|1\rangle$ with equal probability (50–50) upon measurement. Now apply the Hadamard gate a second time, and you will observe that the state of the qubit is represented by an arrow pointing to the north pole of the Bloch sphere. This indicates that the value of the qubit has returned to $|0\rangle$, thereby demonstrating that quantum gates are reversible.

Though we have only seen the effect of the Hadamard gate on a single qubit so far, I would like to make an important statement about the working of quantum gates that provides deep insight into how quantum algorithms operate: all quantum gates correspond to unitary matrix operations acting on state vectors, while measurement is a non-unitary operation. I know that this may sound like another intimidating statement. However, let me repeat my promise: in